SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

DEPT OF EE (VLSI DESIGN & TECH) & DEPARTMENT OF ECE

OFFICIAL HACKATHON DIRECTIVE // 2026 EDITION

VELTRAXX 2.0

National-Level 2-Stage Tech & Silicon Hackathon

PROBLEM STATEMENTS MASTER COMPENDIUM

The authoritative catalog of 25 industrial-grade engineering problem statements across ASIC Design, FPGA Acceleration, Edge AI, Cyber-Physical Systems, and Open Hardware Innovation.

TOTAL STATEMENTS **25 Problem Statements**	DOMAIN TRACKS **5 Industrial Tracks**	EVALUATION TIERS **Tier A (Jr UG) · Tier B (Sr UG/PG)**
ROUND 1 FREE REG **Till 01 Dec 2026 (11:59 PM)**	PPT BLUEPRINT WINDOW **02 Dec to 03 Dec 2026**	24H OFFLINE FINALE **15 Dec to 16 Dec 2026**

EXECUTIVE FRAMEWORK

THE THREE BUILT-IN ANTI-CHEAT ARCHITECTURE RULES

To convert 1,000+ engineers across disciplines while rendering copy-paste submissions ineffective, every problem statement in VELTRAXX 2.0 is governed by three rigorous technical parameters:

RULE 01

Broad Hook, Modular Core

Round 1 requires the foundational architecture, modular block diagram, and concrete Slide 5 Proof of Competence. On December 15, the jury injects a Day-1 Live Twist (bus width changes, parity/fault injection, or sensor failover) that builds directly on your modular core without breaking it.

RULE 02

Dual-Tier Evaluation

Tier A (1st & 2nd Year UG - Junior Division): Graded on architectural logic, functional simulation (Icarus / EDA Playground / Proteus / Wokwi / Python), and working flow.

Tier B (3rd & 4th Year UG, PG & Scholars - Senior Division): Graded on RTL synthesis / PPA reports, timing/power closure, and edge hardware deployment.

RULE 03

Branch Inclusivity

Tracks 1 and 2 cater to hardcore VLSI, Microelectronics, ECE, and Instrumentation builders. Tracks 3 and 4 engage CSE, IT, AI & DS, EEE, and Mechatronics engineers. Track 5 acts as an open multidisciplinary innovation magnet.

MASTER DIRECTORY

TABLE OF TRACKS & PROBLEM STATEMENTS

Track Code	Track Name	Problem Code Range	Statements
VLSI-A	Core ASIC & Custom Silicon Architecture	VTX-T1-01 to VTX-T1-05	**5 Problems**
VLSI-B	FPGA Acceleration & Reconfigurable Hardware	VTX-T2-01 to VTX-T2-05	**5 Problems**
AI/CSE	Edge AI, TinyML & Computer Vision	VTX-T3-01 to VTX-T3-05	**5 Problems**
CSE/ECE/EEE	Smart Systems, Cyber-Physical IoT & Automation	VTX-T4-01 to VTX-T4-05	**5 Problems**
SIH FORMAT	Open Silicon & Hardware Innovation	VTX-T5-01 to VTX-T5-05	**5 Problems**

TRACK 01 // VLSI-A

Target Branches: VLSI, ECE, EEE, Instrumentation

Core ASIC & Custom Silicon Architecture

RTL design, low-power architectures, memory controllers, and DFT/BIST verification systems.

VTX-T1-01

NATIONAL LEVEL INDUSTRIAL SPEC

Low-Power Configurable ALU with Autonomous Logic BIST (LBIST)

The Challenge: Design a synthesizable multi-operation Arithmetic Logic Unit paired with an on-chip Built-In Self-Test (LBIST) controller using an LFSR test-pattern generator and MISR signature analyzer to detect stuck-at faults without external ATE equipment.

TIER A (1ST / 2ND YR UG)

8-bit/16-bit Verilog RTL simulation with basic LFSR pattern generation and self-checking testbench waveforms.

TIER B (3RD / 4TH YR UG & PG)

32-bit pipelined RTL with dual-MISR fault-aliasing reduction, clock-gating for idle power reduction, and synthesis PPA (Power, Performance, Area) reports.

VTX-T1-02

NATIONAL LEVEL INDUSTRIAL SPEC

Metastability-Hardened Dual-Clock Asynchronous FIFO Bridge

The Challenge: Architect a configurable asynchronous FIFO buffer capable of transferring high-speed burst data safely across two independent, unrelated clock domains (CDC) without data loss or metastability failure.

TIER A (1ST / 2ND YR UG)

Behavioral/RTL simulation demonstrating Gray-code pointer conversion, 2-stage flip-flop synchronizers, and accurate full/empty flag generation.

TIER B (3RD / 4TH YR UG & PG)

Parameterized bus/depth RTL, programmable almost-full/almost-empty watermarks, formal CDC verification/STA timing constraints, and synthesis sign-off.

VTX-T1-03

NATIONAL LEVEL INDUSTRIAL SPEC

Fault-Tolerant 5-Port Network-on-Chip (NoC) Mesh Router

The Challenge: Design a 5-port (North, South, East, West, Local) packet-switched NoC router utilizing deterministic XY routing and round-robin arbitration, equipped with single-error detection and recovery against silent data corruption.

TIER A (1ST / 2ND YR UG)

RTL design and simulation of a 5-port crossbar switch with XY routing logic and basic parity-bit error flagging.

TIER B (3RD / 4TH YR UG & PG)

Virtual-channel buffering, Hamming SEC-DED (Single Error Correction, Double Error Detection) datapath protection, latency/throughput benchmarking, and physical floorplan estimation.

VTX-T1-04

NATIONAL LEVEL INDUSTRIAL SPEC

Dynamic Clock-Gated Multi-Master APB/AXI-Lite Peripheral Interconnect

The Challenge: Build an AMBA-compliant bus bridge connecting a high-speed processor master to multiple low-speed peripheral slaves (Timer, GPIO, UART, SPI), automatically gating the clock of idle slaves to minimize dynamic power dissipation.

TIER A (1ST / 2ND YR UG)

RTL implementation of a 1-Master, 3-Slave APB bridge with clean read/write handshaking and functional waveforms.

TIER B (3RD / 4TH YR UG & PG)

AXI4-Lite to APB protocol conversion with glitch-free integrated clock-gating (ICG) cells, timeout watchdog recovery, and synthesized gate-level power analysis.

VTX-T1-05

NATIONAL LEVEL INDUSTRIAL SPEC

Side-Channel Resistant Cryptographic Datapath (Masked AES / Lightweight Cipher)

The Challenge: Edge silicon is vulnerable to Differential Power Analysis (DPA) where attackers extract secret keys from power spikes. Design a hardware encryption engine (PRESENT, Simon, or AES S-Box) that masks intermediate states to flatten power signatures.

TIER A (1ST / 2ND YR UG)

Synthesizable RTL of a lightweight block cipher with a self-checking encryption/decryption testbench.

TIER B (3RD / 4TH YR UG & PG)

First-order Boolean/Domain-Oriented Masking (DOM) implementation with pipelined register barriers and switching-activity comparison reports.

TRACK 02 // VLSI-B

Target Branches: ECE, VLSI, EEE, CSE (Hardware/Systems)

FPGA Acceleration & Reconfigurable Hardware

Hardware accelerators, high-speed streaming interfaces, DSP pipelines, and FPGA prototyping.

VTX-T2-01

NATIONAL LEVEL INDUSTRIAL SPEC

Real-Time 2D Sobel & Canny Edge-Detection Streaming Pipeline

The Challenge: Software image processing introduces frame lag in autonomous drones. Design a line-buffered streaming hardware accelerator in RTL/FPGA that computes spatial gradients on incoming pixel streams in real time with zero frame-buffer external memory stalls.

TIER A (1ST / 2ND YR UG)

Verilog/VHDL simulation processing a test image matrix (Hex/COE file) through a 3x3 sliding window filter and outputting edge-detected image data.

TIER B (3RD / 4TH YR UG & PG)

Full streaming pipeline with dual-port BRAM line buffers, non-maximum suppression, resource utilization (LUT/DSP/BRAM) optimization, and live VGA/HDMI or UART frame output.

VTX-T2-02

NATIONAL LEVEL INDUSTRIAL SPEC

Custom RISC-V (RV32I) Core with Hardware AI/DSP Instruction Extension

The Challenge: Design a 32-bit RISC-V processor core and extend its Instruction Set Architecture (ISA) with custom hardware instructions (such as single-cycle Multiply-Accumulate MAC or ReLU activation) to accelerate edge neural network layers.

TIER A (1ST / 2ND YR UG)

Single-cycle or 3-stage pipelined RV32I base core executing standard arithmetic/branch assembly instructions with a custom MAC opcode in simulation.

TIER B (3RD / 4TH YR UG & PG)

5-stage pipelined RV32I core with data-forwarding, hazard detection, AMBA peripheral bus, and cycle-count benchmark comparison proving speedup over software-only execution.

VTX-T2-03

NATIONAL LEVEL INDUSTRIAL SPEC

Hardware-Accelerated Network Packet Firewall & Deep Pattern Matcher

The Challenge: High-speed routers cannot inspect packets in software without dropping throughput. Architect a wire-speed RTL packet parser that inspects incoming Ethernet/IPv4/UDP headers and blocks malicious payload signatures in a single clock cycle per byte.

TIER A (1ST / 2ND YR UG)

FSM-based header parser extracting Source IP, Destination Port, and Protocol fields from simulated packet streams and triggering a drop flag on blacklisted rules.

TIER B (3RD / 4TH YR UG & PG)

Parallel Bloom-filter or TCAM-style rule lookup engine supporting dynamic rule updates and gigabit-rate timing closure on FPGA architecture.

VTX-T2-04

NATIONAL LEVEL INDUSTRIAL SPEC

Low-Power MFCC Audio Feature Extractor for Always-On Voice Wake-Word

The Challenge: Design a dedicated hardware digital signal processing (DSP) pipeline that converts raw digital microphone audio (I2S/PCM) into spectral features (FFT / Mel-Frequency bands) to trigger a wake-word flag at ultra-low power.

TIER A (1ST / 2ND YR UG)

Fixed-point FIR pre-emphasis filter and windowed energy/zero-crossing detector in RTL simulation capable of distinguishing speech from silence.

TIER B (3RD / 4TH YR UG & PG)

Radix-2 FFT butterfly engine with Mel-filterbank LUT accumulation, fixed-point quantization error analysis, and FPGA DSP-slice optimization.

VTX-T2-05

NATIONAL LEVEL INDUSTRIAL SPEC

Multi-Channel High-Resolution PWM & Dead-Time Motor Drive Controller

The Challenge: Electric vehicle (EV) inverters and robotic actuators require nanosecond-accurate PWM signals with programmable dead-time insertion and sub-microsecond over-current fault shutoff in hardware.

TIER A (1ST / 2ND YR UG)

4-channel programmable PWM generator in RTL with configurable duty cycle, frequency prescaler, and emergency hardware brake pin.

TIER B (3RD / 4TH YR UG & PG)

Complementary 3-phase sinusoidal/SVPWM generator with programmable dead-time counters, rotary encoder quadrature decoder feedback loop, and hardware PID calculation engine.

TRACK 03 // AI/CSE

Target Branches: CSE, AI & DS, IT, ECE, Robotics

Edge AI, TinyML & Computer Vision

TinyML, hardware-aware neural networks, smart diagnostics, and real-time edge computer vision.

VTX-T3-01

NATIONAL LEVEL INDUSTRIAL SPEC

Sub-50ms Quantized Edge Vision System for PCB & Wafer Defect Triage

The Challenge: Cloud-based visual inspection is too slow and expensive for high-speed electronics assembly lines. Build an offline Edge AI vision pipeline capable of detecting missing components, solder bridges, or surface scratches in under 50ms per frame.

TIER A (1ST / 2ND YR UG)

Trained lightweight vision model (YOLOv8-Nano / MobileNet) with a working local UI dashboard displaying bounding-box defect classification and confidence scores.

TIER B (3RD / 4TH YR UG & PG)

INT8 quantized model (TFLite / ONNX Runtime) running under strict memory/CPU caps, handling motion blur or lighting shifts, with automated pass/reject hardware GPIO trigger output.

VTX-T3-02

NATIONAL LEVEL INDUSTRIAL SPEC

Non-Invasive Driver Drowsiness, Gaze & Micro-Sleep Edge Telemetry

The Challenge: Design a real-time in-cabin safety system that tracks eye-aspect ratio (EAR), head pose, and yawning frequency without cloud connectivity, triggering graduated physical alerts before a highway collision occurs.

TIER A (1ST / 2ND YR UG)

Real-time webcam/edge-camera inference pipeline accurately tracking facial landmarks and triggering visual/audio alerts when eye-closure exceeds a threshold time.

TIER B (3RD / 4TH YR UG & PG)

Multi-modal sensor fusion combining camera gaze tracking with steering/accelerometer jitter analysis, low-light IR resilience, and false-positive suppression.

VTX-T3-03

NATIONAL LEVEL INDUSTRIAL SPEC

Acoustic Anomaly & Predictive Bearing Fault Detector for Factory Motors

The Challenge: Industrial motors emit subtle ultrasonic and acoustic frequency shifts days before catastrophic mechanical failure. Build a TinyML/Edge-DSP diagnostic engine that classifies normal vs. anomalous machine states from raw audio/vibration streams.

TIER A (1ST / 2ND YR UG)

Spectrogram/FFT feature extraction pipeline paired with a lightweight classifier that identifies faulty vs. healthy motor audio samples in real time.

TIER B (3RD / 4TH YR UG & PG)

On-device continuous anomaly detection capable of filtering out background factory noise, running on a microcontroller/edge board with live health-index telemetry.

VTX-T3-04

NATIONAL LEVEL INDUSTRIAL SPEC

Offline Edge-LLM & RAG Assistant for Lab Equipment & Datasheet Triage

The Challenge: Engineers debugging hardware in secure labs often lack internet access or waste hours digging through 500-page component datasheets. Build a 100% offline, local Small Language Model (SLM) + Retrieval-Augmented Generation (RAG) system that extracts exact pinouts, timing limits, and register maps with zero hallucination.

TIER A (1ST / 2ND YR UG)

Working local RAG pipeline that ingests PDF datasheets and answers technical queries with exact page/table citations.

TIER B (3RD / 4TH YR UG & PG)

Quantized local model running purely on laptop CPU/NPU (zero external API calls), featuring automated truth-table/schematic parameter extraction and strict anti-hallucination guardrails.

VTX-T3-05

NATIONAL LEVEL INDUSTRIAL SPEC

Precision Agricultural Drone Imagery Triage for Early Crop Stress

The Challenge: Drones capture thousands of high-resolution field images, but farmers need instant localized action maps before landing. Build an ultra-lightweight vision classifier that segments leaf blight, pest damage, and water stress at the edge.

TIER A (1ST / 2ND YR UG)

Image classification and segmentation pipeline identifying crop disease categories with severity percentage estimation and geo-tagged report generation.

TIER B (3RD / 4TH YR UG & PG)

Pruned/quantized edge model optimized for low-power flight compute, calculating targeted pesticide/water dosage maps in real time at 15+ FPS.

TRACK 04 // CSE/ECE/EEE

Target Branches: ECE, EEE, CSE, IT, Mechatronics, Biomedical

Smart Systems, Cyber-Physical IoT & Automation

Highway safety, industrial telemetry, smart BMS automation, and integrated sensor-cloud systems.

VTX-T4-01

NATIONAL LEVEL INDUSTRIAL SPEC

Non-Linear Sensor Noise Rejection & Closed-Loop Actuator Stabilization

The Challenge: Ultrasonic, ToF, and IR sensors suffer from severe impulse spikes and environmental multipath noise, causing robotic servos and industrial valves to jitter violently. Build a real-time digital filtering and closed-loop control system with live telemetry plotting.

TIER A (1ST / 2ND YR UG)

Microcontroller prototype (Arduino/ESP32) reading noisy distance/environmental sensors, applying moving-average/median filtering, and driving a smooth actuator response.

TIER B (3RD / 4TH YR UG & PG)

Adaptive Kalman or non-linear median-PID control loop rejecting deliberate impulse noise injection, paired with a real-time 60fps serial telemetry dashboard plotting raw vs. stabilized signals.

VTX-T4-02

NATIONAL LEVEL INDUSTRIAL SPEC

EV Battery Management System (BMS) with Thermal Runaway & Cell Balancing

The Challenge: Lithium-ion battery packs fail when individual cells overcharge or overheat. Architect a smart BMS monitoring node that tracks per-cell voltage, current, and temperature gradients, isolating faulted cells in milliseconds.

TIER A (1ST / 2ND YR UG)

Multi-sensor monitoring prototype (or multi-channel hardware simulation) tracking voltage/temperature thresholds with automated relay/MOSFET cutoff and LCD/Web status alerts.

TIER B (3RD / 4TH YR UG & PG)

State-of-Charge (SoC) & State-of-Health (SoH) estimation algorithm, active/passive cell-balancing logic, predictive thermal-runaway rate-of-rise (dT/dt) detection, and CAN/MQTT telemetry.

VTX-T4-03

NATIONAL LEVEL INDUSTRIAL SPEC

Decentralized V2X Highway Black-Spot & Fog Collision Avoidance Mesh

The Challenge: On foggy highways and blind mountain hairpin bends without 4G/5G coverage, vehicles need instant peer-to-peer hazard warnings. Build a low-latency local wireless mesh (ESP-NOW / LoRa / RF / BLE) that broadcasts sudden braking, black-ice, or obstacle alerts between nodes.

TIER A (1ST / 2ND YR UG)

Two or more hardware nodes communicating hazard/distance states peer-to-peer without internet router dependency, triggering instant visual/buzzer warnings.

TIER B (3RD / 4TH YR UG & PG)

Multi-hop packet relay with time-to-live (TTL) deduplication, RSSI/ToF proximity estimation, NFC/cryptographic node authentication, and live command-center map visualization.

VTX-T4-04

NATIONAL LEVEL INDUSTRIAL SPEC

Smart Grid Micro-Load Shedding & Power-Theft Tamper Detection Node

The Challenge: Distribution transformers overload during peak hours and suffer revenue loss from illegal line tapping. Build an intelligent smart-energy node that detects differential current tampering and dynamically sheds non-critical loads before the main breaker trips.

TIER A (1ST / 2ND YR UG)

Smart energy meter prototype measuring load current/power, detecting overload conditions, and automatically switching priority vs. non-priority relay channels.

TIER B (3RD / 4TH YR UG & PG)

Dual-point current differential tamper detection, power-factor/harmonic anomaly tracking, automated load-rebalancing state machine, and tamper-proof cloud/local audit logging.

VTX-T4-05

NATIONAL LEVEL INDUSTRIAL SPEC

Wearable Multi-Vital Patient Triage Node with Fall & Arrhythmia Alerting

The Challenge: General hospital wards lack continuous ICU-grade monitoring for every bed. Build a low-power wearable telemetry patch/node that monitors heart-rate variability, SpO2, skin temperature, and sudden falls, alerting nurses immediately upon clinical deterioration.

TIER A (1ST / 2ND YR UG)

Sensor prototype (MAX30102 / MPU6050 / temperature) acquiring live vitals, filtering motion artifacts, and displaying real-time alerts on a nurse-station web dashboard.

TIER B (3RD / 4TH YR UG & PG)

On-device Early Warning Score (EWS) state machine, store-and-forward offline buffer when Wi-Fi drops, ultra-low-power sleep scheduling, and multi-patient priority triage sorting.

TRACK 05 // SIH FORMAT

Target Branches: Open to All Engineering Domains (UG, PG & Research Scholars)

Open Silicon & Hardware Innovation

Pitch your own bold hardware or silicon-backed solution to a real-world industrial or societal problem.

VTX-T5-01

NATIONAL LEVEL INDUSTRIAL SPEC

Open Innovation in Assistive Tech & Neuro-Rehabilitation Hardware

The Challenge: Pitch and build any original hardware, embedded, or VLSI-backed assistive device empowering differently-abled individuals (speech-to-sign, smart prosthetics, eye-tracked wheelchair control, or tactile navigation).

TIER A (1ST / 2ND YR UG)

Working prototype or simulation demonstrating functional assistive feedback loop.

TIER B (3RD / 4TH YR UG & PG)

Engineered hardware device with clinical/ergonomic metrics, low-latency actuation, and edge power optimization.

VTX-T5-02

NATIONAL LEVEL INDUSTRIAL SPEC

Open Innovation in Sustainable Tech, Water & Climate Hardware

The Challenge: Pitch and build an original sensor-driven, IoT, or custom silicon accelerator system addressing water leakage/contamination, micro-climate forecasting, solar MPPT optimization, or e-waste sorting.

TIER A (1ST / 2ND YR UG)

Functional sensor ingestion node with calibrated telemetry and local action threshold alerting.

TIER B (3RD / 4TH YR UG & PG)

Multi-sensor closed-loop control system, predictive environmental analytics, and field deployment reliability testing.

VTX-T5-03

NATIONAL LEVEL INDUSTRIAL SPEC

Open Innovation in Defense, Aerospace & Secure Communications

The Challenge: Pitch and build an original hardware/software system for anti-drone detection, GPS-denied navigation, encrypted tactical comms, or radiation/fault-tolerant electronics.

TIER A (1ST / 2ND YR UG)

Simulated or breadboard communication prototype demonstrating secure peer-to-peer data transport.

TIER B (3RD / 4TH YR UG & PG)

Ruggedized embedded node with cryptographic authentication, frequency-hopping/mesh resilience, and hardware fault-recovery.

VTX-T5-04

NATIONAL LEVEL INDUSTRIAL SPEC

Open Innovation in EDA Automation, Scripting & Verification Tooling

The Challenge: Pitch and build a developer tool, Perl/TCL/Python automation script, automated testbench generator, waveform debugger, or linting engine that speeds up the IC/hardware design lifecycle.

TIER A (1ST / 2ND YR UG)	TIER B (3RD / 4TH YR UG & PG)
Automated script or tool capable of parsing Verilog/VHDL code and detecting syntax/latching errors.	Comprehensive EDA workflow accelerator with automated test vector synthesis, CDC violation checks, or GUI waveform visualization.

VTX-T5-05

NATIONAL LEVEL INDUSTRIAL SPEC

Wildcard Disruptive Silicon & Cyber-Physical Systems

The Challenge: Have a bold hardware, FPGA, ASIC, robotics, or edge-compute architecture that does not fit into the categories above? Pitch your blueprint, prove its engineering feasibility on Slide 5, and bring it to the 24-hour arena.

TIER A (1ST / 2ND YR UG)	TIER B (3RD / 4TH YR UG & PG)
End-to-end architectural proposal with block diagram, component specification, and initial simulation/mockup.	Full hardware/RTL prototype with measured benchmarks, real-world stress test results, and production feasibility roadmap.

SUBMISSION PROTOCOL

ROUND 1 BLUEPRINT GUIDELINES & CONTACT DIRECTORY

Mandatory 6-Slide Presentation Specification

All teams must download the official PowerPoint template from `veltraxx.in/downloads`, complete the 6 slides, export as a PDF under 3MB, and upload via the submission portal between **02 Dec (00:00) and 03 Dec (23:59)**:

- **Slide 1:** Project Identity & Engineering Team Roster (2-4 Members)
- **Slide 2:** Problem Deconstruction, Failure Mechanism & Numerical Targets
- **Slide 3:** Proposed Microarchitecture & Step-by-Step Data Flow Pipeline
- **Slide 4:** Tools, Hardware Boards, Core Algorithms & Validation Strategy
- **Slide 5 (CRITICAL):** Concrete Proof-of-Competence Artifact (Waveform, Truth Table, State Transition Table, or Register Map)
- **Slide 6:** 24-Hour Arena Milestone Roadmap & Quantified Real-World Impact

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